



Use of machine learning to achieve keratoconus detection skills of a corneal expert

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Received: 28 November 2021 / Accepted: 13 June 2022 / Published online: 11 August 2022
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Abstract

Purpose To construct an automatic machine-learning derived algorithm discriminating between normal corneas and suspect irregular or keratoconic corneas.

Methods A total of 8526 corneal tomography images of 4904 eyes obtained between November 2010 and July 2017 using a combined Scheimpflug/Placido tomographer were retrospectively evaluated. Each image was evaluated for acquisition quality and was labeled as normal, suspect irregular or keratoconic by a cornea specialist.

Two algorithms were built. The first was based on 94 instrument-derived output parameters, and the second integrated keratoconus prediction indices of the device with the 94 instrument-derived output parameters. Both models were compared with the tomographer's keratoconus detection algorithms. Out of the 8526 images evaluated, 7104 images of 3787

eyes had sufficient acquisition quality. Of those, 5904 examinations were randomly chosen for construction of the models using the random forest algorithm. The models were then validated using the remaining 1200 examinations.

Results Both RF algorithms had a larger AUC compared with any of the tomographer's KC detection algorithms ($p < 10^{-9}$). The first constructed model had 90.2% accuracy, sensitivity of 94.2%, and specificity of 89.6% (Youden 0.838). Calculated AUC was 0.964. The second model had 91.5% accuracy, sensitivity of 94.7%, and specificity of 89.8% (Youden 0.846). Calculated AUC was 0.969.

Conclusion Using the RF machine-learning algorithm, accuracy of discrimination between normal, suspect irregular and keratoconic corneas approximates that of an experienced corneal expert. Applying machine learning to corneal tomography can facilitate keratoconus screening in large populations as well as off-site screening of refractive surgery candidates.

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Keywords Machine learning · Artificial intelligence · Random forest · Keratoconus · Detection · Galilei · Dual Scheimpflug/Placido

Introduction

Keratoconus (KC) is a bilateral progressive non-inflammatory disorder characterized by ectasia and thinning of the cornea. The progressive thinning

and subsequent anterior protrusion of the cornea can result in severe myopic astigmatism and asymmetric irregular astigmatism, leading to distorted vision [1]. KC is a common disorder, with prevalence rates fluctuating geographically between 0.17 in 1000 in the USA [2] to 40 in 1000 in Iran [3]. KC prevalence among 662,644 Israeli adolescents was found to be 10 in 1000 [4]. A similar clinical picture of corneal ectasia can occur as a postoperative complication of laser refractive surgery [5].

The strong correlation between keratoconus and systemic disorders such as atopic dermatitis, obstructive sleep apnea, connective tissue diseases and down syndrome has been previously described [6–8] as well as a strong correlation with a wide range of ocular disorders like floppy eyelid syndrome, allergic conjunctivitis, corneal and retinal dystrophies [8].

A treatment option for mild to moderate KC in recent years is collagen cross linking (CXL) which can slow or even halt disease progression [9]. It is therefore important to implement effective screening programs to detect the disease in its early stages, so that CXL can be employed to prevent further progression. KC diagnosis at its advanced stages is less challenging compared to earlier stages of the disease where it can be overlooked. Published literature includes dozens of indices used to aid in the detection of KC [10].

The Galilei tomographer (Ziemer, Port, Switzerland) has a dual Scheimpflug rotating camera and integrated Placido disc. Corneal Placido and Scheimpflug images are obtained simultaneously in a single scan and are used for anterior and posterior corneal measurements. The Galilei produces 94 measurement parameters for every scan and has several software-integrated algorithms for keratoconus detection.

Machine learning algorithms have been used previously for classification tasks in different domains, with a growing use of such algorithms in various medical fields. Neural networks have been in use since 1995 on data extracted from corneal topographies to detect corneal pathologies [11]. Other algorithms evolved, mainly discriminating between normal and KC corneas, and lately detecting forme fruste KC. More sophisticated methods of machine learning algorithms were later utilized including support vector analysis (SVM) [12, 13], decision trees [14],

random forest [15] and picture analyzer techniques [16, 17].

Due to the high prevalence of KC, development of accessible screening methods is warranted. Shortage of ophthalmologist in general and cornea specialists specifically, enhances the need for an automatic screening system. A study published by Resnikoff et al. in 2015 estimating the number of ophthalmologist worldwide, depicts high variability between low income countries vs high income countries for the number of ophthalmologist per capita, ranging between 3.7 per million to 76 per million, respectively [18].

The aim of this study was to investigate whether a random forest (RF) machine learning algorithm constructed using a very large database can approximate the ability of an experienced corneal specialist to discriminate between normal, suspect irregular and frank KC corneal images.

Patients and methods

This retrospective study was conducted at the Tel Aviv Sourasky Medical Center (Tel Aviv, Israel) in association with the School of Electrical Engineering at the Tel Aviv University (Tel Aviv, Israel). The study was approved by the Institutional Review Board of the Tel Aviv Sourasky Medical Center and was conducted in accordance with the tenets of the Declaration of Helsinki.

A total of 8526 corneal tomography examinations of 2525 patients obtained between November 2010 and July 2017 with the Galilei dual Scheimpflug/Placido disc analyzer system (software version 5.2.1; Ziemer Ophthalmic Systems, Port, Switzerland) were evaluated. Examination output was in a six-map format: Axial Curvature (anterior), Elevation with best fit sphere (BFS) (anterior and posterior), Elevation with best fit toric asphere (BFTA) (anterior and posterior), and pachymetry. For each image, the relevant maximum and minimum values were presented. All images were graded by a single cornea specialist (D.V.) according to the six-map format without knowing patient demographics or any other patient data. Each examination was graded for acquisition quality and classified for the presence of keratoconus. Acquisition quality grading was divided into low-quality scan and high-quality scan based on corneal

coverage, missing data points, quality of fixation and lid artifacts and was made exclusively by the discretion of the corneal expert. Keratoconus classification was 1=normal, 2=suspect irregular cornea and 3=keratoconus. A normal cornea would have a regular spherical or spherocylindrical curvature, thinning toward the center without epicentral posterior or anterior elevation, with relatively normal numerical values. A suspected irregular cornea describes an at-risk cornea. Such a cornea may have subtle inconclusive signs like inferior-superior values outside the reference range or aberrant C-shaped or mild posterior surface elevations. Alternatively, a suspected irregular cornea may have an unusual corneal thinning.

Output format examples of images classified as normal (Fig. 1), suspect irregular cornea (Fig. 2) and keratoconus (Fig. 3) are presented.

Of the 8526 evaluated samples, 1422 samples were graded as low quality and were excluded from the analysis, leaving 7104 samples that were included in

the study. Of the 7104 included samples (3787 eyes of 2384 patients), 1200 samples were randomly selected, using the Scikit-learn software version 1.0.2, to be later used as a validation set. The remaining 5904 samples were used first to construct the model.

Of the 7104 included samples, 4088 were labeled as normal, 1299 were labeled as suspect irregular cornea, and 2614 were labeled as keratoconus. Label distribution was similar in train and test sets (3508 and 580 labeled as normal, 1099 and 200 as suspect irregular cornea, and 2194 and 420 as Keratoconus, respectively).

Two models were constructed. The first model used 94 variables measured by the device. Following construction of the model, the 1200-sample validation set was used to validate the model's diagnostic ability. The model was then compared with the tomographer's previously validated KC detection algorithms [19], including: keratoconus prediction index (KPI), keratoconus probability (Kprob), cone location and

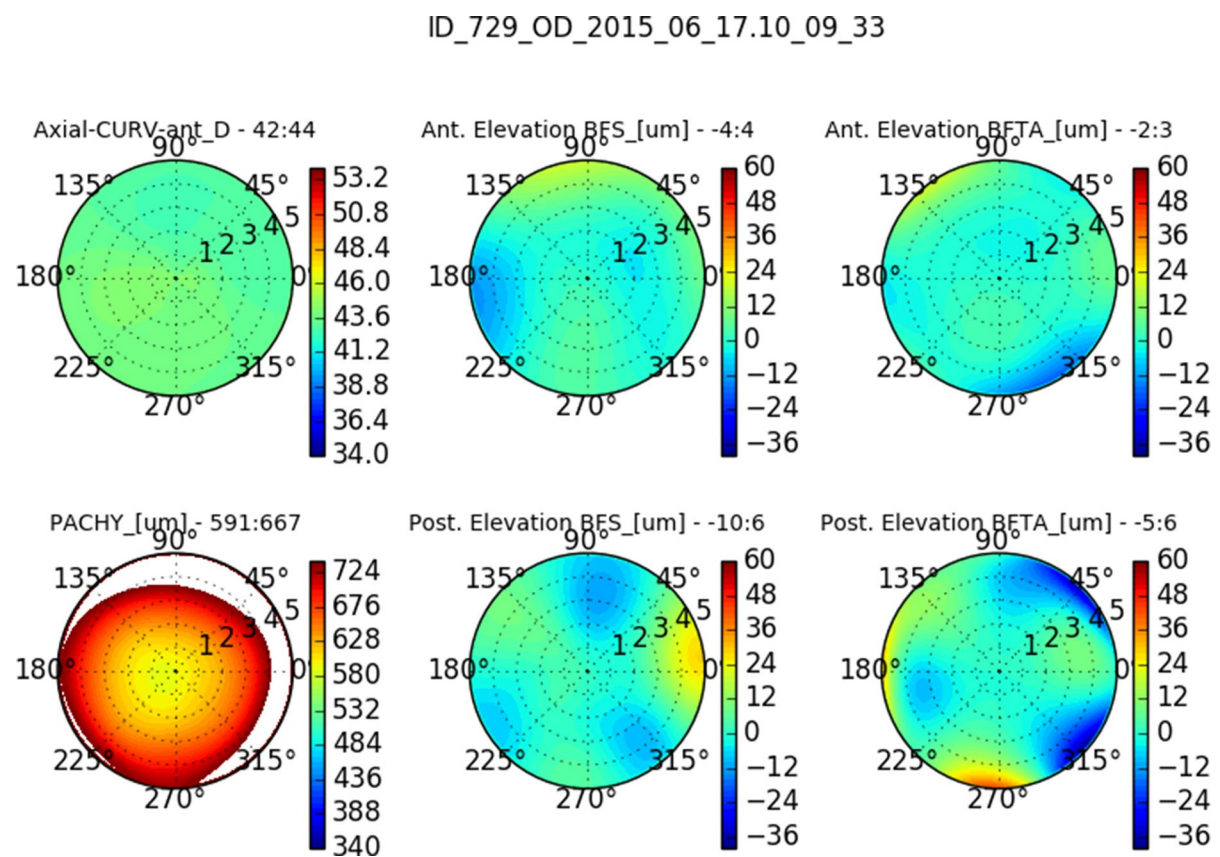


Fig. 1 Representative Dual Scheimpflug/Placido image categorized as normal

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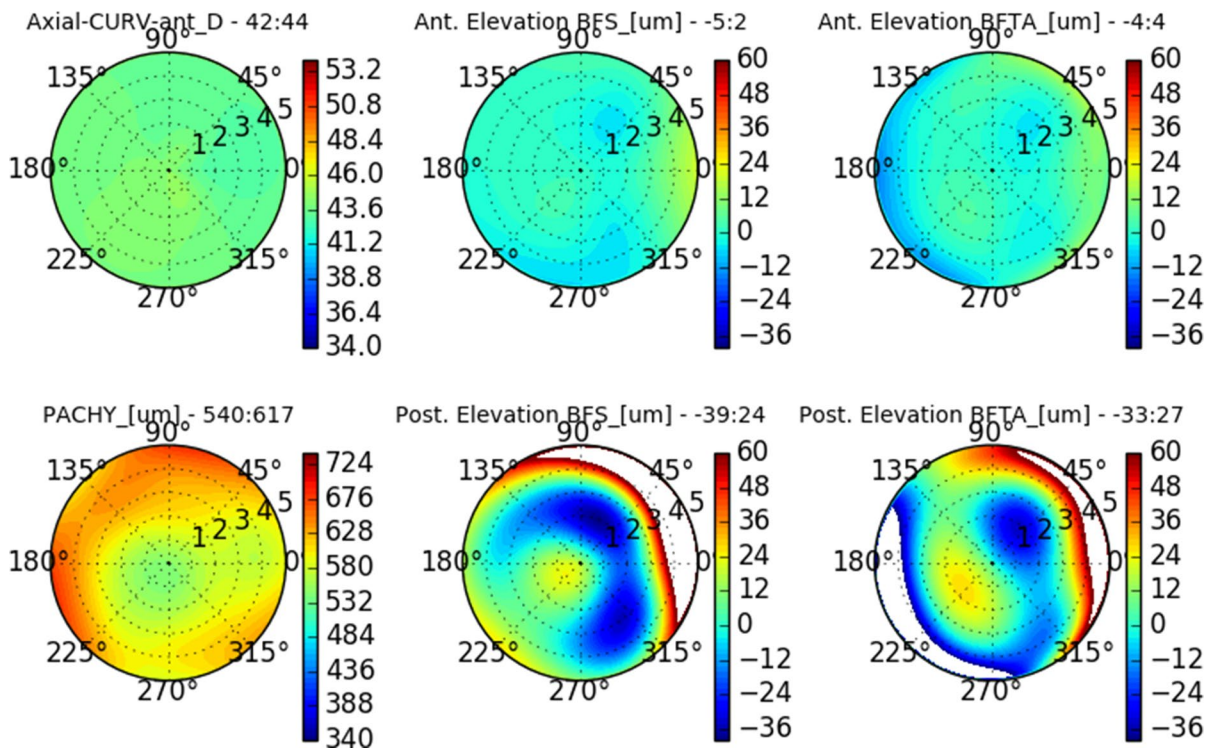


Fig. 2 Representative Dual Scheimpflug/Placido image categorized as suspect irregular cornea

magnitude index (CLMI), percentage probability of keratoconus (PPK), inferior-superior index (IS), differential sector index (DSI), surface asymmetry index (SAI), opposite sector index (OSI), surface regularity index (SRI), center/surround index (CSI), irregular astigmatism index (IAI), and standard deviation of corneal power (SDP).

The second model integrated keratoconus (KC) prediction indexes of the device in addition to the 94 instrument-derived output parameters. The same 1200-sample validation set was used to validate the second model's diagnostic ability.

Random forest algorithm

Decision trees (DT) are a popular method for various machine learning tasks [20]. A decision tree splits the data into separate regions (leaf nodes), each one with its own prediction. For regression, it is common to predict

the mean of the training data in that region for prediction, and the majority label for classification. If a probability is needed, it is possible to output the portion of each label out of the total samples in the region. For example, if a region consists of 40 healthy corneas and 60 corneas with Keratoconus, then any cornea which arrives to that region would be predicted with a 60% chance of having Keratoconus. From that probability, it is easy to gain the certainty of the model.

The tree is built by recursively splitting the data by one of the features, such that the loss function is minimized. For regression, the common loss function is the residual sum of squares (RSS):

$$\text{RSS} = \sum_i (y_i - \hat{y}_i)^2,$$

where i refers to the sample number, y_i to its label, and $\hat{y}_i$ to its prediction.

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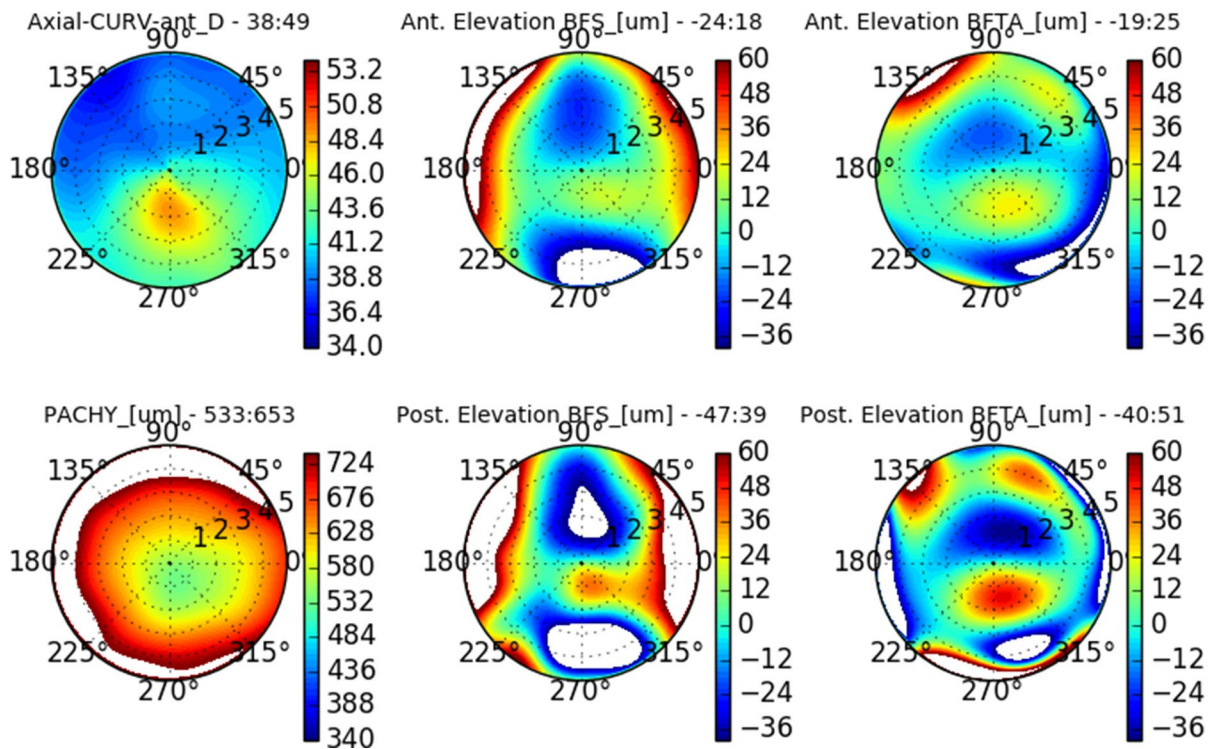


Fig. 3 Representative Dual Scheimpflug/Placido image categorized as keratoconus

For classification, the common loss function is the Gini index [21]:

$$\text{Gini Index} = 1 - \sum_i (p_i)^2$$

where i refers to the sample number, and p_i to the fraction of its label in its leaf.

Though intuitive (the decision process is clear to the user), a single tree usually overfits the training data—which means that it fits well to the training data, but fails to generalize [22]. Various algorithms use more than one tree in their process to overcome this problem. The model used in the current study was the random forest model.

In this model, we aggregate the predictions of multiple trees to generate the final prediction. The decision trees differ by 2 randomization processes:

1. Bootstrap sample—The data set of each tree consists of n sampled samples from the dataset with repetition.
2. Feature randomization—At each split of each tree, only a subset of the features are considered as candidates for partition. Usually, the subset size is the square root of the features number.

To gain intuition of the importance of a feature, it is possible to average its impact (loss of RSS\ Gini index) across all trees, by observing the nodes where it was used.

This method was used to see which are the most important features that the model used.

Statistical analysis

Statistical analysis was performed using the R 3.6.2 for windows software 2019 (R Core Team, Vienna, Austria).

After performing a verification for a normal distribution for the entire group and for both learning set and validation set separately, the AUC of the random forest model was compared to AUCs of the tomographer's integrated KC detection algorithms by calculating a two-sided *P*-value [23, 24] where the Kendall tau was used for correlation analysis, and the variance of the Wilcoxon was used for the AUC standard error analysis.

Receiver operating characteristics (ROC) curves were used to assess the discriminating ability and to determine the optimal cutoff values for each significant metric and multivariable combinations.

In all analyses, a *P* value less than 0.01 was considered statistically significant.

Accuracy was computed by the mean of the predictions score, where a prediction which was equal to its examination classification yielded a score of 1, and otherwise 0.

Results

The first RF algorithm, constructed using the 94 device-measured parameters, had 90.2% accuracy with a sensitivity of 94.2% and specificity of 89.6%

(Youden 0.838) on the 1200 validation samples. Calculated AUC was 0.964.

The second RF algorithm, constructed using the tomographer's KC detection algorithms in addition to the 94 device-measured parameters, had 91.5% accuracy with a sensitivity of 94.7% and specificity of 89.8% (Youden 0.846) on the 1200 validation samples. Calculated AUC was 0.969. Table 1 shows an AUC and accuracy comparison between the tomographer's KC detection algorithms and the RF algorithms. Both RF algorithms had a significantly larger AUC compared with any of the

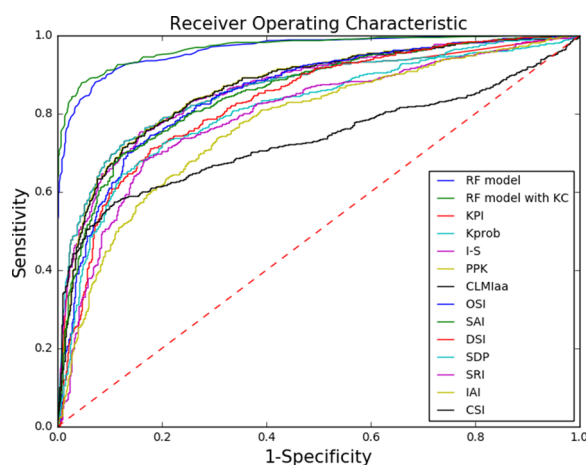


Fig. 4 ROC Curve—comparison between the different parameters for keratoconus detection and the RF model

Table 1 Comparison between the tomographer's KC detection algorithms and the RF algorithms by AUC and accuracy

Predictor	AUC	Accuracy (%)	<i>P</i> -Value*
Random Forest without KC algorithm	0.964	90.2	
Random Forest with KC algorithm	0.969	91.5	
Keratoconus Prediction Index (KPI)	0.782	75.0	< 10 ⁻⁹
Keratoconus probability (Kprob)	0.774	73.1	< 10 ⁻⁹
Inferior—Superior index (IS)	0.783	71.4	< 10 ⁻⁹
Percentage Probability of Keratoconus (PPK)	0.772	71.1	< 10 ⁻⁹
Cone Location and Magnitude index (CLMI)	0.777	71.8	< 10 ⁻⁹
Opposite Sector Index (OSI)	0.769	70.5	< 10 ⁻⁹
Surface Asymmetry Index (SAI)	0.765	70.2	< 10 ⁻⁹
Siderential Sector Index (DSI)	0.749	68.3	< 10 ⁻⁹
Standard Deviation of corneal Power (SDP)	0.737	68.2	< 10 ⁻⁹
Surface Regularity Index (SRI)	0.686	63.6	< 10 ⁻⁹
Irregular Astigmatism Index (IAI)	0.641	59.3	< 10 ⁻⁹
Center/Surround Index (CSI)	0.693	62.9	< 10 ⁻⁹

**p* value for AUC comparing both RF models with the tomographer's KC detection algorithms

tomographer's KC detection algorithms ($p < 10^{-9}$ for all comparisons, Table 1).

Figure 4 shows the ROC curve comparison between the tomographer's KC algorithms and the RF models.

Figures 5 and 6 specify the ten most influential measured variables in decision-making for the two

RF models. Each feature is measured for its importance, in the presence of the others.

In the first model, 9 out of the 10 most important measurements used were measurements of the anterior cornea and of corneal thickness (Fig. 5).

In the second model, 6 out of the 10 most important parameters that were used for constructing the

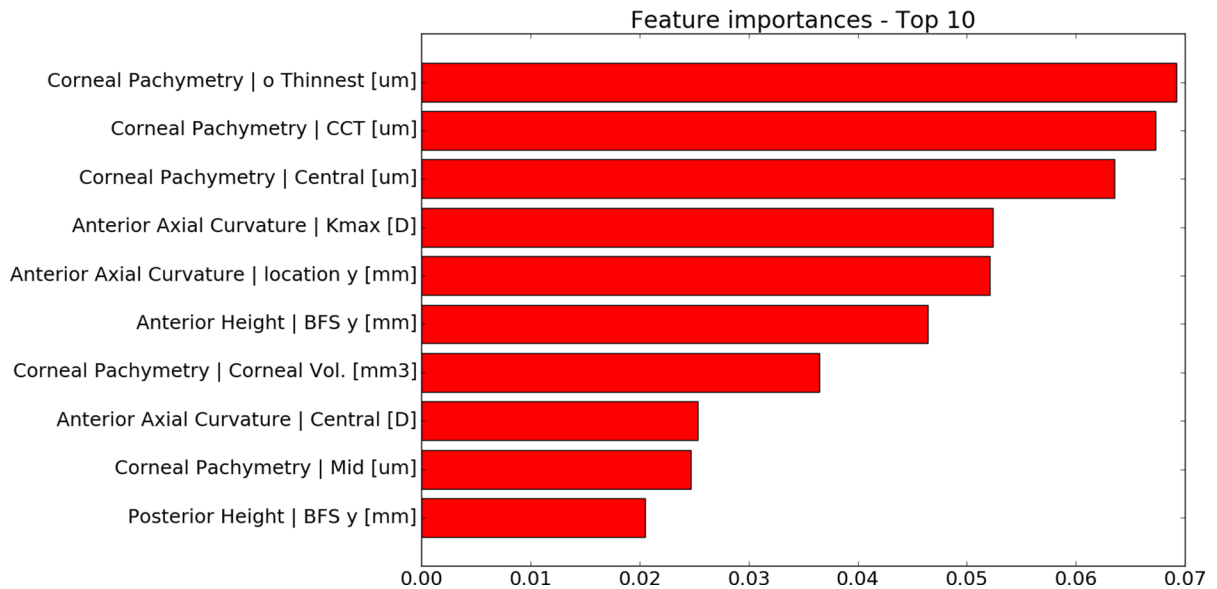


Fig. 5 Ten most influential measured variables in RF model decision-making without tomographer's integrated KC algorithm

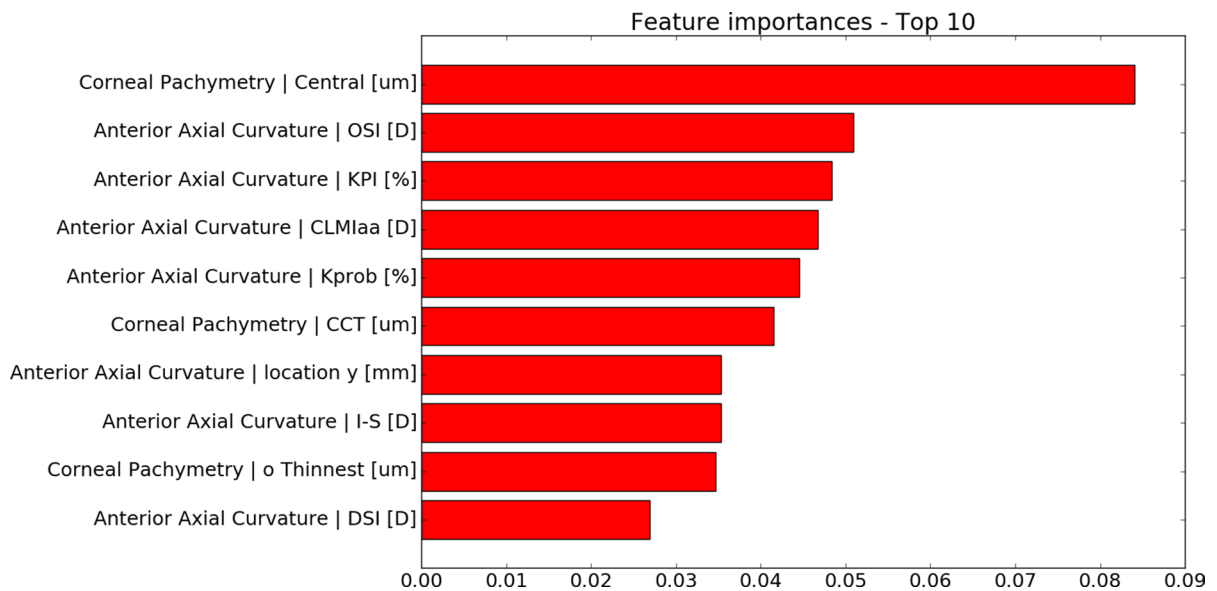


Fig. 6 Ten most influential measured variables in RF model decision-making with tomographer's integrated KC algorithm

model were part of the tomographer's KC detection algorithms. The rest were measurements of the anterior cornea and corneal thickness (Fig. 6).

Discussion

Machine learning is an artificial intelligence technique which uses a computerized system to perform a specific task without being given explicit instructions. The machine constructs a mathematical algorithm based on a case dataset of known inputs and results which trains the machine to mimic human learning abilities. Machine learning has been employed vastly in several ophthalmology subfields including retina (diabetic retinopathy screening [25]), glaucoma (retinal nerve fiber analysis and visual field interpretation [26, 27]) and neuro ophthalmology [28].

Maeda and Smolek were among the first to employ machine learning algorithms for automated corneal pattern recognition using corneal topographies [29–31]. Souza et al. were the first to use tomographic images obtained by the Orbscan II for automated discrimination between KC and non-KC images. They used three artificial intelligence methods: radial basis function neural network, support vector machine and multi-layer perception, showing an AUC of 0.99, 0.99 and 0.98, respectively. They concluded that all three machine learning techniques showed high efficacy discriminating between KC and non-KC images [16]. Smadja et al. were the first to use Galilei images to create an automated decision tree classifier differentiating between normal, form fruste KC (FFKC) and frank KC, using a 55-indices algorithm which was employed on 372 tomographic maps, with sensitivity/specificity of 100%/99.5% in discriminating normal vs KC and 93.6%/97.2% in discriminating normal vs FFKC. The most important discriminating variables in their model were posterior surface asymmetry index and corneal volume [14]. It should be noted that these studies had smaller cohorts with cross-validation for determining accuracy.

Recently, Golan et al. employed a machine learning algorithm distinguishing 178 normal post-LASIK corneas from clinically unaffected eyes with highly asymmetric keratoconus (FFKC) using 87 metrics parameters obtained from the Galilei device and showed that no single metric achieved an AUC greater than 0.79, but a combined model consisting of

9 metrics yielded an AUC of 0.96 with 90.3% sensitivity and 92.6% specificity. Five of those metrics were related to corneal pachymetry [32]. In our first model, five out of the ten most important parameters were corneal pachymetry-related, and in the second model, three out of the ten most important parameters were corneal pachymetry-related. Central pachymetry was the most influential parameter for constructing the model. This finding coincides with the global consensus on keratoconus and ectatic diseases published in 2015 by a group of panelists for the global delphi panel of keratoconus and ectatic diseases. The panel agreed that abnormal corneal thickness distribution and abnormal posterior ectasia are mandatory for the diagnosis of keratoconus [33].

In the current study, posterior corneal parameters were found to have the least important role in construction of both our models, as shown in Figs. 5 and 6. Interestingly, Hwang et al. created a model based on 16 indices obtained from Scheimpflug imaging (Pentacam HR, Oculus, Inc., Arlington, WA) and SD OCT (RTVue-100, Optovue, Inc., Fremont, CA) for discriminating normal corneas from FFKC corneas using multivariate logistic regression. Ninety topographic maps were evaluated with the model reaching an AUC of 1.0 with 100% sensitivity and specificity. This study concluded that posterior corneal metrics were not significantly impactful [34].

Keratoconus, in the majority of the cases, shows a displacement of the corneal apex with localized areas of steepening over the vertical meridian of the cornea, the corneal apex tends to displace inferotemporally to create a vertical asymmetry in the corneal power and as expected in our first model, three out of ten most influence parameters concern displacement over the “y” axis.

Lopes et al. published their study on post-LASIK ectasia, evaluating 3693 Pentacam tomography maps. Their model differentiated between normal eyes and frank post-LASIK ectasia eyes as well as between normal eyes and normal fellow eyes of ectatic cases. Results were AUC of 1.0 and 0.968, respectively [15]. Xie et al. lately published their study on Asian patient database, distinguishing corneas with contraindications for refractive surgery, with an overall 6465 corneal tomographic images of 1385 patients. Their model was based on Pentacam HR images and Pentacam InceptionResNetV2 Screening System (PIRSS) artificial intelligence (AI) model to compare accuracy

between the AI and human diagnoses. Their model had an overall accuracy of 94.7% on the validation set. The PRISS model performed better than the classifiers in the Pentacam HR system. These findings are similar to the current study where the RF model had better prediction than all other integrated KC detection algorithms of the device. Last, Chen et al. [35] published their study on keratoconus detection of changes using deep learning of color-coded maps; in this study, they showed an accuracy of 0.9785 of the convolutional neural networks technique for discriminating between normal and keratoconus corneas by using a concatenated four maps image (axial map, anterior and posterior elevation map and pachymetry map); compared to their results, we had a lower accuracy although we used a six-map images and this is most probably due to the fact that our model also had a suspect irregular cornea group that the model had to take into account.

Our study uses the largest patient cohort to date and separates the eyes on which the model was trained from the eyes on which it was tested. The models described in this study were aimed at distinguishing normal corneas from at risk corneas and frank KC corneas. Although the terms normal and frank KC corneas are more consensual, the term at risk cornea varies in the literature and is often being described as suspect KC which can be manifested as high K values, high asymmetry indices or subtle anterior or posterior elevation maps which do not fall within the frank KC category. Another term being used is forme fruste KC (FFKC) which usually refers to mild irregularities not evolving to keratoconus, sometimes in a fellow eye of a clinically diagnosed KC eye. Subclinical KC is another term being used for describing at risk corneas with no abnormal clinical finding, typically seen in KC, on a slit lamp examination. In our model, we used the term suspect irregular cornea to define at risk cornea with suspicious corneal morphologic abnormalities.

Our study has several limitations. The model relies on one surgeon categorizing the images to each group. The created model is therefore aiming to mimic the categorization made by that surgeon. In the absence of a gold standard for suspect irregular corneas, we argue that mimicking the real-life experience of one surgeon may be superior to using the tomographer's multiple integrated KC algorithms, often disagreeing with each other. A second limitation is that our model

does not take into consideration age, sex or ethnical parameters which are known to have an effect on corneal morphology [36, 37]. Finally, the model is based upon morphologic characteristics only with no biomechanical properties taken into consideration.

For future studies adding a non-subjective parameters as the orientation of the apical displacement and morphological aberrometric analysis may increase the sensitivity and specificity of our model and have to be investigated as an added values.

In conclusion, a screening tool based on Galilei Scheimpflug/Placido tomography and a machine-learning algorithm that automatically differentiates between normal corneas to suspect irregular and keratoconic corneas was constructed using basic variables routinely measured by the tomographer. This tool's diagnostic ability was superior to any of the tomographer's pre-existing KC detection algorithms. Integration of this algorithm into the device's software could serve as an adjunct KC diagnosis tool. Applying machine learning to corneal tomography can facilitate keratoconus screening in large populations as well as off-site screening of refractive surgery candidates.

Authors' contribution All authors contributed to the study conception and design. Material preparation, data collection and analysis were performed by DV, EC, DB, and NS. The first draft of the manuscript was written by EC, and all authors commented on previous versions of the manuscript. All authors read and approved the final manuscript.

Funding All authors declare that no grant support or research funding was received for the purpose of this study. All authors certify that they have no affiliations with or involvement in any organization or entity with any financial interest (such as honoraria; educational grants; participation in speakers' bureaus; membership, employment, consultancies, stock ownership, or other equity interest; and expert testimony or patent-licensing arrangements), or non-financial interest (such as personal or professional relationships, affiliations, knowledge or beliefs) in the subject matter or materials discussed in this manuscript. This article does not contain any studies with human participants or animals performed by any of the authors.

Declarations

Conflict of interest The authors of this manuscript has no competing interests to declare.

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